

Evaluation of MPPT Algorithms for PV System under Partial Shading Conditions

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Abstract— The harmful impacts of fossil fuels have made us adopt renewable energy sources which do not replenish upon more usage. The renewable energy sources like solar PV system provide many benefits, including green and clean energy. At the same time, solar PV panels suffer from low output power when subjected to partial shading. Several Maximum Power Point Tracking (MPPT) algorithms have been proposed in the literature to tackle this problem. Choosing an appropriate algorithm to track the maximum power point is of paramount importance as it improves the efficiency of the PV system by a significant margin. This paper focuses on evaluating and comparing various MPPT algorithms that fall under different categories such as conventional, soft computing and hybrid methods. These algorithms are simulated and tested on a partially shaded PV system for various performance parameters in MATLAB/Simulink environment.

Keywords— MPPT algorithm, PV system, partial shading, boost converter, optimization, hybrid method.

I. INTRODUCTION

The usage of fossil fuels has led to some adverse effects today. Some of them are harmful impacts on the environment and a shortage of resources in the future. Governments worldwide have been encouraging renewable energy to avoid depleting the resources for future generations and safeguard our planet. The renewables such as the solar PV and the wind turbines have made way for green and clean energy. Along with various advantages, the solar PV system has its downside which is the low conversion efficiency. Hence, extracting the maximum power from the panels becomes essential by using a suitable MPPT controller. The PV System exhibits PV curves with single MPP when exposed to uniform shading but shows multiple peaks when subjected to non-uniform shading. Partial shading leads to a dramatic reduction in energy production. The effect of partial shading has a detrimental effect which can be seen in small-scale to the large-scale PV power plant. Partial shading can lead to elevated temperature of the modules and can also create hot spots. These hot spots lead to cracks in the PV modules which can degrade the total power generated by a significant amount. An extensive analysis of the effect of partial shading on PV array is carried out in [1].

To track the global maxima during shading, many MPPT techniques have been proposed in the literature [2,3]. MPPT algorithms are broadly classified as conventional methods, soft computing techniques, and hybrid methods [4,5]. In [6] a

concise classification and evaluation review of all the applied MPPT methods has been presented. Conventional methods can easily track MPP under uniform shading conditions but fail under partial shading. Perturb and Observe (PO), Hill climbing and Incremental Conductance (INC) are some examples of this category [7]. Soft computing techniques are complex but are highly efficient even during partial shading [8,9]. Such technique mainly involves Artificial Intelligence (AI) or evolutionary algorithms. The AI method can easily track the MPP even under partial shading conditions but involves more complexity. Fuzzy logic control and optimization methods are some of the AI-based techniques. Among the above two approaches, the optimization method is less complex. Popular Evolutionary algorithms are Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), Ant colony, and Artificial Bee Colony algorithms [10]. The hybrid methods try to combine both conventional and AI-based techniques to overcome the disadvantages of each of them. Hybrid methods could also be a combination of two conventional or two soft computing techniques. In [11], the $0.8 V_{oc}$ model and an adaptive scanning procedure is used in hybrid combination to lessen the scanning time interval. GWO as in [12] and PO have been fused to improve the convergence time in [13]. In [14] the PSO technique is used in a hybrid combination with PO. Eight different Global Maximum Power Point Techniques (GMPPT) are assessed in [15] but the study does not include any hybrid methods. The execution of the optimization-based MPPT algorithm is assessed and compared using simulation tools in terms of tracking speed, accuracy and oscillation in output power at steady state in [16]. In [17], the optimization methods are classified as fixed and variable, and are simulated and compared. To find the best suited MPPT method for a specific application, it is necessary to evaluate and compare the existing methods. This paper attempts to simulate and compare MPPT techniques from all three categories of conventional, soft computing, and hybrid methods. Four different MPPT techniques are simulated in MATLAB/Simulink. The various methods are compared in terms of precision, power fluctuations, speed of convergence. A DC-DC boost converter is designed for the MPPT application as in [18].

II. DESIGN OF PV SYSTEM FOR SHADED CONDITION

A. PV system model

A bypass diode will be connected across a group of PV cells, bypassing the shaded module and allowing the

unshaded modules to continue to supply power as shown in Fig 1(a). During partial shading, this configuration results in several peaks on the PV curve. Fig. 1(b) and Fig. 1(c)

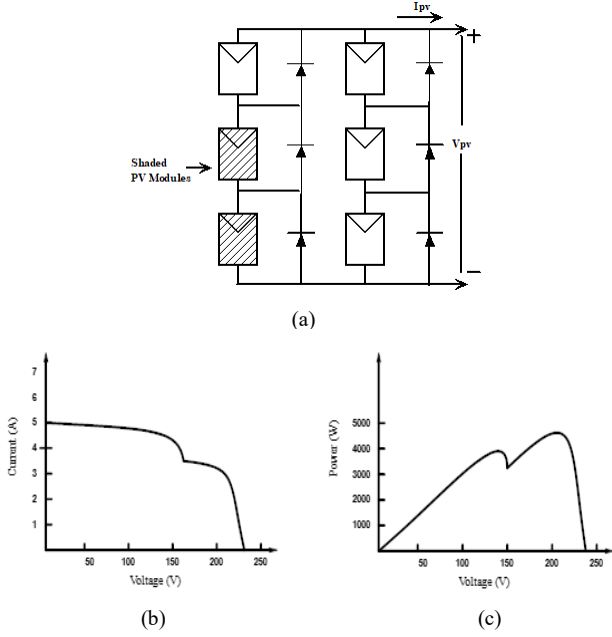


Fig. 1. a) Partial shading of PV array (b) I- V (c) P-V curves of PV array during partial shading

depicts the I-V and P-V curves under shaded conditions. As illustrated in Fig. 1(c), there is only one GMPP amid the many peaks.

$$I_{SC} = I_{SCREF} \times [1 + \alpha(T - T_{REF})] \times \left(\frac{S}{S_{REF}}\right) \quad (1)$$

$$V_{OC} = V_{OCREF} \times n \times \left[1 + \alpha \ln\left(\frac{S}{S_{REF}}\right) + \beta(T - T_{REF})\right] \quad (2)$$

$$I = I_{REF} \times \frac{I_{SC}}{I_{SCREF}} \quad (3)$$

$$I_{REF} = I_{SCREF} \times 1 - k_{ref} \frac{V - V_{OC} + (R_S \times I_{REF} \times \frac{I_{SC}}{I_{SCREF}})}{V_{OCREF} \times n} \quad (4)$$

The authors of [20], propose a model for simulation of PV array which aids to explore the characteristics of a PV array under varying condition of insolation and temperature. The simulation model can assist in foreseeing the performance of MPPT both in uniform and non-uniform irradiance. With reference to the PV model from [20] the mathematical expression for current and voltage are given by (1), (2), (3) and (4). The panel output current I is given in terms of I_{REF} (Current at STP), OC voltage V_{OC} , SC current I_{SC} , and constant k_{REF} using the PV array specifications shown in Table 1. Where α and β are the temperature coefficient of I_{SC} and V_{OC} respectively.

B. Design of boost converter

The boost converter must be designed to account for the PV system's fluctuating solar insolation. The inductance and capacitance are calculated in [18] with the nonlinear nature of the PV source in mind. The maximum and minimum values of the PV panel resistance, are used to compute the

optimum load resistance. The PV system's maximum and minimum resistances are $R_{mp(max)}$ and $R_{mp(min)}$, respectively.

The maximum and minimum values of the duty cycles are D_{max} and D_{min} respectively. The relationship between the output resistance R_O and the PV system resistance R_{mp} is described in (5) and (6). To avoid the discontinuous conduction mode, the inductance L_{min} is computed with the least amount of irradiation possible as in (7). Using the specifications of boost converter from Table 2, the input capacitance C_i and output capacitance C_o are also computed as shown in (8) and (9).

Table 1: Specifications of PV Array

Number of Panels	10
Reference Insolation (S_{REF})	1000 W/m ²
Reference Temperature (T_{REF})	25 C°
Reference SC Current (I_{SCREF})	4.6A
Reference OC Voltage (V_{OCREF})	15 V
Reference MPP Voltage (V_{MPPREF})	10V
Reference MPP Current (I_{MPPREF})	4A
Reference MPP Power (P_{MPPREF})	40 W

Table 2: Specifications of Boost Converter

Input voltage (PV output) V_{mp}	100 V
Output Voltage (V_O)	220 V
Output Power (P_O)	400 W
Inductor (L)	2.32 mH
Capacitor (C_o)	13 μ F
Load Resistance	121 Ω
Current Ripple	30%
Voltage Ripple	1%

$$R_O = \frac{R_{mp}}{(1 - D)^2} \quad (5)$$

$$D = 1 - \sqrt{\frac{R_{mp}}{R_O}} \quad (6)$$

$$L_{min} = \frac{I_L R_{mp(max)}}{\Delta i_L f} \left(1 - \sqrt{\frac{R_{mp(max)}}{R_{O(max)}}}\right) \quad (7)$$

$$C_i = \frac{D}{8L \frac{\Delta V_{mp}}{V_{mp}} f^2} \quad (8)$$

$$C_o = \frac{D(1 - D)^2}{R_{mp} \frac{\Delta V_O}{V_O} f} \quad (9)$$

Fig. 2 depicts the schematic block diagram representation of the PV system under study, which employs a DC-DC boost converter for impedance matching. Direct duty ratio control is employed since it reduces the control circuit's complexity.

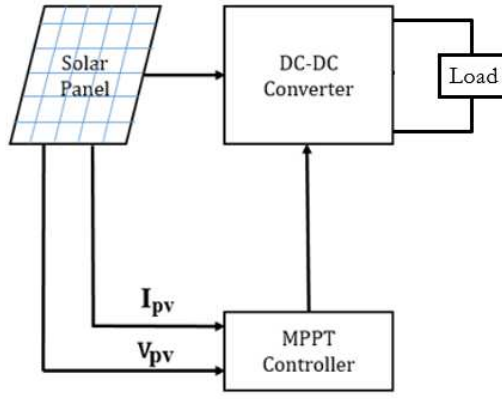


Fig. 2. Schematic block diagram representation of PV system

III. MPPT ALGORITHMS

This section discusses various conventional and optimization strategies employed under both uniform and partial shading conditions. This section also highlights the hybrid methodologies used in the comparison.

A. Conventional MPPT Algorithms

The PO and INC are two common MPPT approaches among the traditional methods [7]. The former is easier to implement, whereas the latter is more precise. The operating point is continuously checked and perturbed in the PO approach until the MPP is reached. For each perturbation step size, PV power is estimated. The operating point falls in the left side of the peak point if the difference in power before and after perturbation is positive; otherwise, it is on the right side of MPP. As a result, the duty cycle is changed to bring the operating point closer to MPP. However, at steady state, this approach causes fluctuations in the power. The step size can be reduced to control the oscillations, although this increases the convergence time. Similarly, the INC method keeps track of the slope of the PV curve. For each sample time, the PV current I and voltage V are monitored. ΔI and ΔV are the differences between the current and prior measurements. The incremental conductance denoted by $\Delta I/\Delta V$, and the conductance I/V are compared. $dP/dV=0$ once the incremental conductance is equal to the negative of the conductance. The MPP is found at this operational point [7].

B. Soft Computing Algorithms

The PSO [19] and GWO algorithms [12] are two popular methods in this field. The best thing about these optimization strategies is that they don't need any previous knowledge of the PV system. PSO is a stochastic evolutionary algorithm. A swarm of particles is sent out in PSO to find the best target. The duty cycles in the MPPT application reflect the particles. Fig. 3 depicts the PSO flow chart. The dc-dc converter is subjected to a random set of duty cycles. PV power is estimated for each of the duty cycles. P_{best} will be the name given to these values. G_{best} refers to the duty cycle that contributes to maximum power. The duty cycles are updated in future iterations until the MPP point is achieved.

C. Hybrid MPPT Algorithms

Soft computing and traditional methodologies could be combined in hybrid algorithms. PSO-PO [14] and PSO-INC

[21] techniques are two popular hybrid combinations. The PSO method tracks the global peak point, while the PO/INC algorithm sustains the maximum power obtained. Fig. 4 and Fig. 5 depict the flow chart of hybrid PSO-PO and PSO-INC respectively. In comparison to other techniques, the convergence time required to track maximum power is less. Another hybrid method is the improved $0.8 V_{oc}$ method [11], which combines the $0.8 V_{oc}$ model with the scanning strategy, as shown in the flow chart of Fig. 6. Different zones or regions are selected based on the total number of peaks possible and the duty cycle range is calculated accordingly. Scanning is performed in the different regions designated by the duty cycle range in this method. Scanning is limited to the selected areas, making the process faster than traditional scanning. The PO algorithm maintains the duty to sustain the global peak after the duty ratio matching to peak power is discovered.

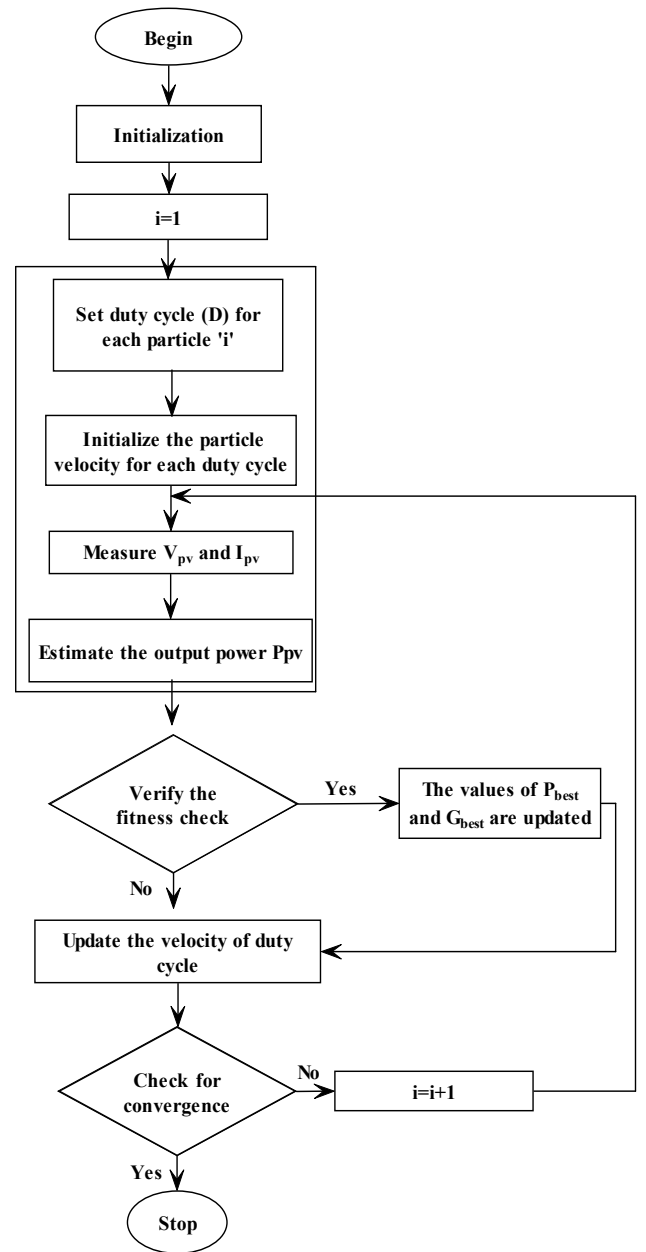


Fig. 3. Flow chart for PSO method [16]

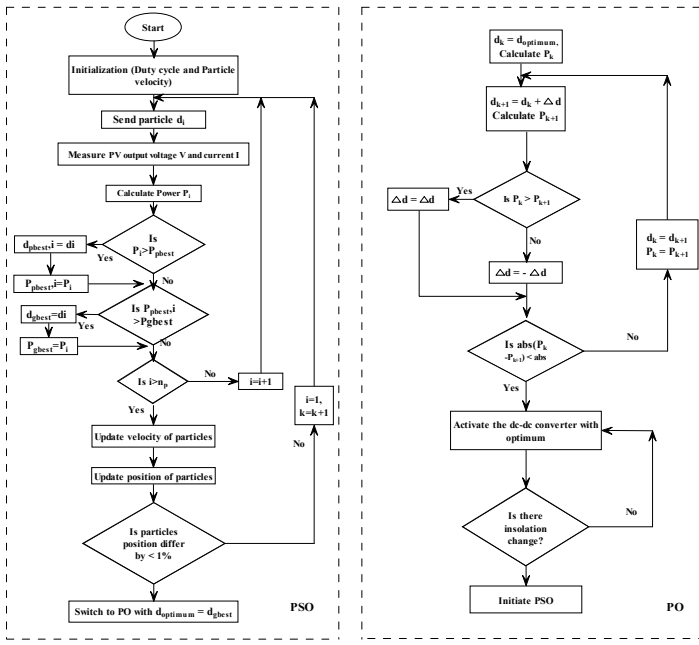


Fig. 4. Flow chart for PSO- PO technique [14]

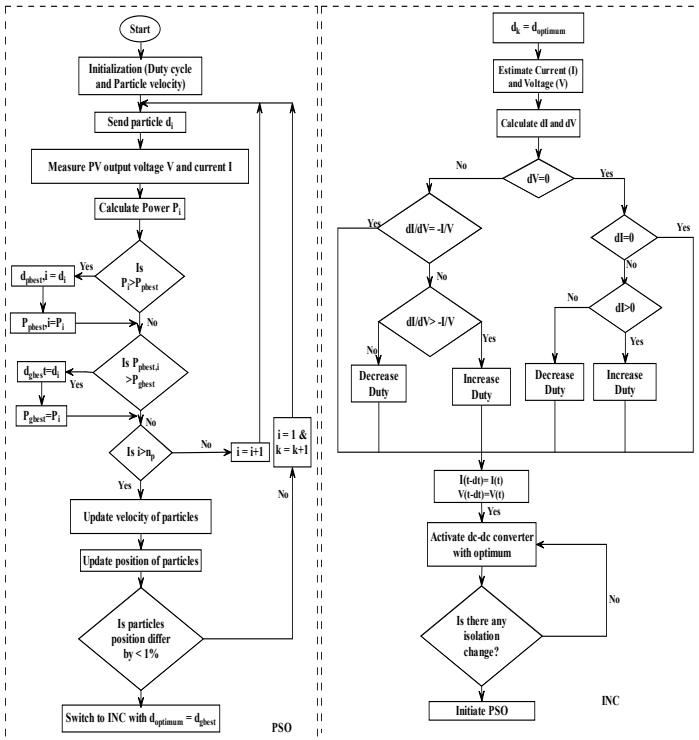


Fig. 5. Flow chart for PSO-INC technique [21]

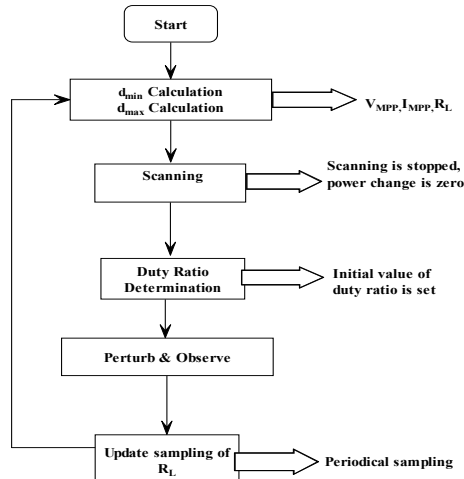


Fig. 6. Flow chart for scanning the curve method [11]

IV. RESULTS AND DISCUSSION

MATLAB Simulink is used to simulate three types of MPPT algorithms mentioned in the previous section. The PV array and boost converter specifications are highlighted in the II section. Both uniform and partial shading are applied to the PV array. A solar irradiance of 1000W/m^2 is maintained for uniform shading. Fig. 7 illustrates the partial shading patterns applied in the simulation.

The traditional PO algorithm was the first algorithm chosen for the investigation. The PV array is shaded uniformly for 1.5 seconds, as seen in Fig. 8. The MPPT controller detects the peak power of 400W, and the settling time is 1 second. Pattern 1 is presented at time $t=1.5$ seconds, and pattern 2 is introduced at time $t=3$ seconds. During shaded conditions, the PO algorithm has a settling time of less than 0.5 seconds and can follow the global peak of 284W for the shading pattern-1 and 200W for shading pattern-2. However, complex shading patterns are known to cause the PO algorithm to fail to follow the global peak [4,5]. The INC approach is the second traditional algorithm evaluated for comparison. As illustrated in Fig. 9, the shading conditions are kept the same as in the preceding scenario. The MPPT controller takes 0.2 seconds to settle for uniform shading and partial shading. In this example study, the INC tracks the global peak, however it will not perform smoothly for a bigger PV array [4,5].

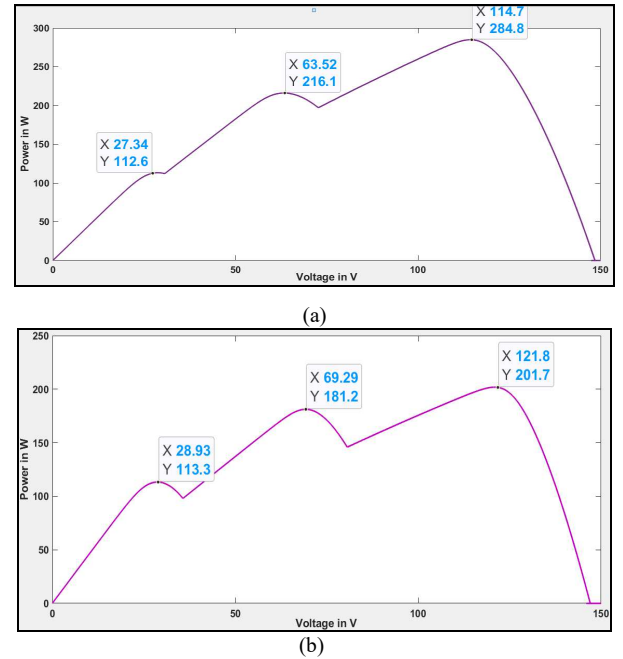


Fig. 7. (a) Partial shading pattern 1 (b) Partial shading pattern 2

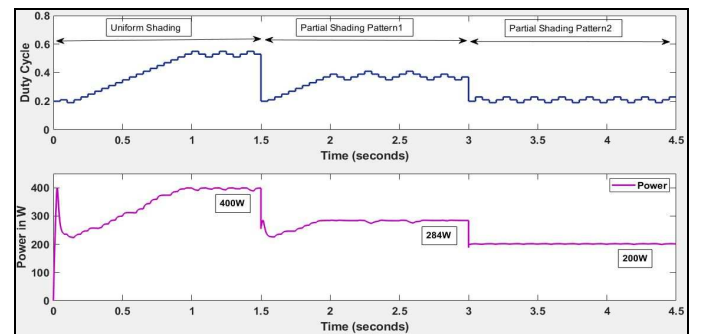


Fig. 8. Variation of duty cycle and output power for PO method

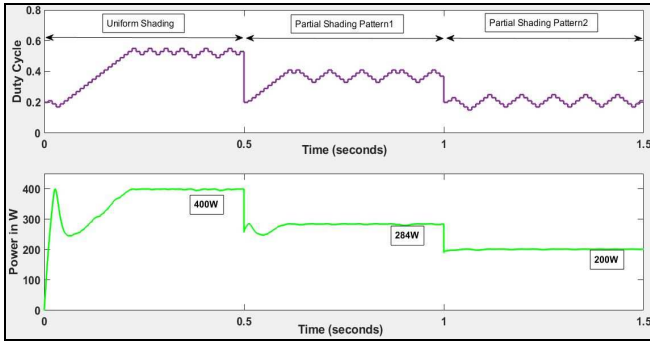


Fig. 9. Variation of duty ratio and output power for INC technique

The PSO is chosen as the next algorithm for the study. As illustrated in Fig. 10, the PSO MPPT tracks the peak in 0.72 seconds for both uniform and partial shading. In a sequential order, three duty cycles are initialised. The PSO approach is more accurate than the conventional method, although it takes longer to convergence. In addition to better accuracy, steady state oscillations are reduced.

The PSO-PO approach is the first hybrid method to be simulated. When uniform shading appears, PO is triggered with settling time of 0.2 seconds. Partial shading necessitates the use of PSO, which obtains the duty cycle for peak power before passing control to PO and settling time is 0.4 seconds. The variation in power and duty cycle is seen in Fig. 11.

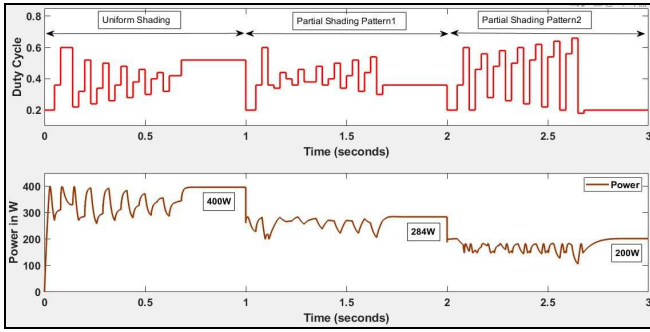


Fig. 10. Variation in duty cycle and output power in PSO method

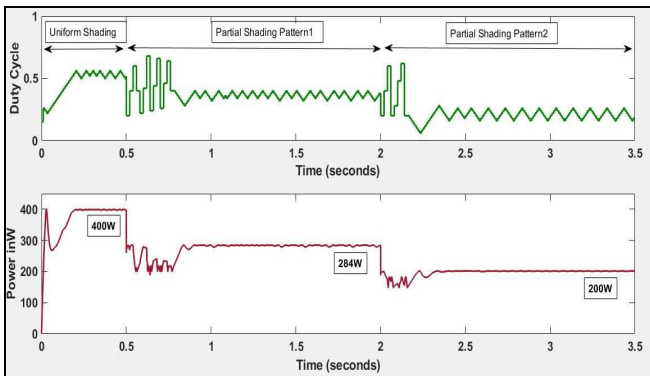


Fig. 11. Variation of duty cycle and power output in PSO-PO method

The PSO-INC methodology is the second hybrid method. Just the INC algorithm is used for uniform shading. In 0.15 seconds, the MPP is tracked. When partial shading is observed, PSO is activated, and iteratively seeks the optimal duty cycle. To keep the peak power, the PO algorithm is fed the optimal duty cycle. In this situation, the settling time is approximately 0.35 seconds as seen in Fig. 12.

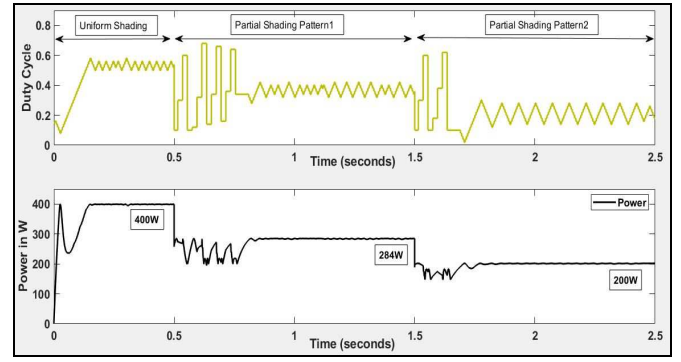
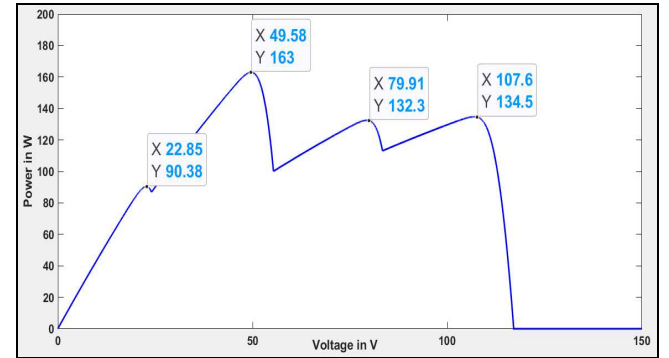
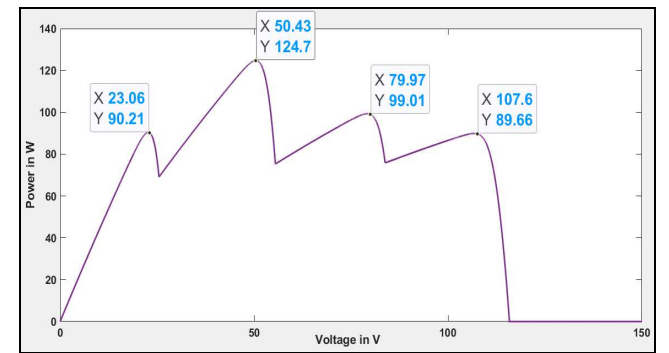


Fig. 12. Variation of duty cycle and power output in PSO-INC method

The third hybrid technique uses a modified $0.8V_{oc}$ model with scanning approach. The minimum and maximum PV resistances $R_{mp(max)}$ and $R_{mp(min)}$ are used to determine the maximum and minimum duty cycles D_{max} and D_{min} . This method's scanning is depicted in Fig. 6. For uniform shading, the PO algorithm is used. When partial shading is seen, the curve is scanned, all peak values are saved, and the global peak is identified. To maintain the maximum power, the duty cycle corresponding to the global peak is transmitted to the PO algorithm. Shading patterns employed for the scanning technique is shown in Fig. 13. The global peak for the first pattern is 163W and for the second is 124.7W. Under uniform shading the maximum power obtained is 400W for a duty cycle of 0.5. Under partial shading, there are 4 peaks in the shading patterns. The algorithm stores the corresponding 4 duty cycles as shown in Fig. 14. At the end of the scanning, the peak power tracked is 160W for the first shading pattern and 120W for the second pattern. Settling time is 0.34 seconds. The obtained power for the duty cycle is plotted in MATLAB and is shown in the Fig. 14



(a)



(b)

Fig. 13. Scanning curve method:(a) Shading pattern 1 (b) Shading pattern 2

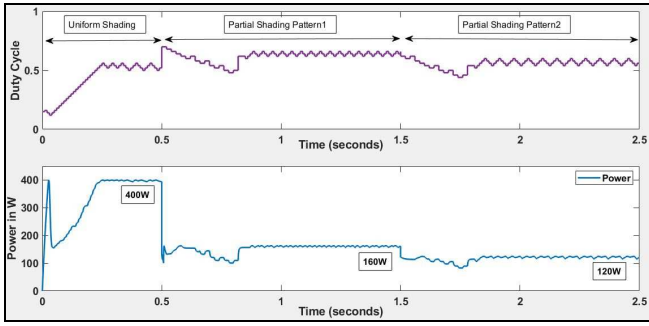


Fig. 14. Variation of duty cycle and power output in scanning approach

The convergence time, steady state oscillations, and accuracy of each of the approach are all compared in Table 3. Depending on which parameter is crucial for a particular application, an appropriate MPPT approach can be chosen. PSO-INC is found to be superior to other methods as it shows less steady state oscillations, low convergence time and higher accuracy.

Table 3. Comparison of the MPPT Techniques

Algorithm	Shading	Power Oscillation	Convergence time	Accuracy	Tracking Speed
PSO	Uniform	0.05%	0.72sec	99%	Medium
	Partial	0.002%	0.72sec	98.5%	Medium
Hybrid PSO-PO	Uniform	0.44%	0.2sec	99.5%	High
	Partial	2.17%	0.4sec	98.59%	Medium
Hybrid PSO-INC	Uniform	0.32%	0.15sec	99.6%	High
	Partial	0.7%	0.35sec	99.29%	High
Scanning the curve	Uniform	0.55%	0.26sec	98.75%	High
	Partial	2.7%	0.34sec	97.93%	Low

V. CONCLUSION

It is challenging to choose a suitable MPPT approach for a shaded PV system because each methodology has its own set of benefits and drawbacks. Comparing the available techniques makes choosing an appropriate MPPT method simple for PV system-based applications. Different MPPT algorithms are tested on a partially shaded PV system for various performance indices in MATLAB/Simulink platform. The results of this simulation study show that PSO method provides high accuracy at the expense of a longer convergence time and a higher level of complexity. The hybrid PSO-PO method reduces the convergence time during partial shading but introduces steady-state oscillations and increases complexity. The PSO-INC hybrid technique is the most accurate method with less convergence time and power oscillations, but it is complex. The scanning the curve approach is a better option in terms of ease of implementation, but it has a lower accuracy.

REFERENCES

- [1] R. Ahmad, A. F. Murtaza, H. A. Sher, U. T. Shami and S. Olalekan, "An analytical approach to study partial shading effects on PV array supported by literature," Elsevier: Renewable and Sustainable Energy Reviews, vol. 74, pp. 721-732, July 2017.
- [2] A. Mohapatra, B. Nayak, P. Das, and K. B. Mohanty, "A review on MPPT techniques of PV system under partial shading condition," Elsevier: Renewable and Sustainable Energy Reviews, vol. 80, pp. 854-867, December 2017.
- [3] M. Mao, L. Cui, Q. Zhang, K. Guo, L. Zhou, and H. Huang, "Classification and summarization of solar photovoltaic MPPT techniques: A review based on traditional and intelligent control

strategies," Elsevier: Energy Reports, vol. 6, pp. 1312-1327, November 2020.

- [4] R. Babu Bollipo, S. Mikkili, and P. Kumar Bonthagorla, "Hybrid, Optimal, Intelligent and Classical PV MPPT Techniques: A Review," CSEE Journal of Power and Energy Systems, vol. 7, no. 1, pp. 9-33, January 2021.
- [5] R. B. Bollipo, S. Mikkili, and P. K. Bonthagorla, "Critical Review on PV MPPT Techniques: Classical, Intelligent and Optimisation," IET Renewable Power Generation, vol. 14, no. 9, pp. 1433-1452, February 2020
- [6] A. O. Baba, G. Liu and X. Chen, "Classification and Evaluation Review of Maximum Power Point Tracking Methods," Elsevier, Sustainable Futures, vol. 2, pp. 100020, April 2020.
- [7] K. Jain, M. Gupta, and A. K. Bohre, "Implementation and Comparative Analysis of P&O and INC MPPT Method for PV System," IEEE India International Conference on Power Electronics (IICPE), May 2018.
- [8] K. Y. Yap, C. R. Sarimuthu, and J. M. Lim "Artificial Intelligence Based MPPT Techniques for Solar Power System: A review," Journal of Modern Power Systems and Clean Energy, vol. 8, Issue: 6, pp 1043-1059, November 2020.
- [9] H. Rezk, M. Al-Oran, M. R. Gomaa, M. A. Tolba, A. Fathy, M. Ali Abdelkareem, A. G. Olabi, and A. H. M. El-Sayed, "A novel statistical performance evaluation of most modern optimization based global MPPT techniques for partially shaded PV system," Elsevier: Renewable and Sustainable Energy Reviews, vol 115, pp. 109372 November 2019.
- [10] F. Belhachat and C. Larbes, "A review of global maximum power point tracking techniques of photovoltaic system under partial shading conditions," Elsevier: Renewable and Sustainable Energy Reviews, vol. 92, pp. 513-553, September 2018.
- [11] M. E. Başoğlu and B. Çakır, "Hybrid global maximum power point tracking approach for photovoltaic power optimisers," IET Renewable Power Generation, vol. 12, no. 8, pp. 875-882, March 2018.
- [12] S. Mohanty, B. Subudhi and P. R. Kumar, "A New MPPT Design Using Grey Wolf Optimization Technique for Photovoltaic System Under Partial Shading Conditions," IEEE Transactions on Sustainable Energy, vol. 7, no. 1, pp. 181-188, January 2016.
- [13] S. Mohanty, B. Subudhi and P. R. Kumar, "A Grey Wolf-Assisted Perturb & Observe MPPT Algorithm for a PV System," IEEE Transactions on Energy Conversion, vol. 32, no. 1, pp. 340-347, March 2017.
- [14] K. Sundareswara, V. Kumar and S. Palani, "Application of a combined particle swarm optimization and perturb and observe method for MPPT in PV systems under partial shading conditions," Elsevier: Renewable Energy, vol. 75, pp. 308-317, March 2015.
- [15] C. M. A. Luz, E. M. Vicente, and F. L. Tofoli, "Experimental evaluation of global maximum power point techniques under partial shading conditions," Elsevier: Solar Energy, vol. 196, pp. 49-73, January 2020.
- [16] M. V. Rocha, L. P. Sampaio and S. A. O. Silva "Comparative Analysis of ABC, Bat, GWO and PSO Algorithms for MPPT in PV Systems," IEEE Explore International conference on Renewable Energy Research and Applications, February 2020.
- [17] J. Dadkhah and M. Niroomand "Optimization Methods of MPPT parameters for PV Systems: Review, Classification, and Comparison," Journal of Modern Power Systems and Clean Energy, vol. 9, no. 2, pp. 225-236, March 2021.
- [18] R. Ayop and C. W. Tan, "Design of boost converter based on maximum power point resistance," Elsevier, Solar Energy, vol. 160, pp. 322-335, January 2018.
- [19] K. Ishaque and S. Zainal, "A Deterministic Particle Swarm Optimization Maximum Power Point Tracker for Photovoltaic System Under Partial Shading Condition," IEEE Transactions on Industrial Electronics, vol. 60, no. 8, pp. 3195-3206, August 2013.
- [20] K. Ding, X. Bian, H. Liu, and T. Peng, "A MATLAB-Simulink-Based PV Module Model and Its Application Under Conditions of Nonuniform Irradiance," IEEE Transactions on Energy Conversion, vol. 27, no. 4, pp. 864-872, December, 2012.
- [21] M. Abdulkadir and A. H. M. Yatim, "Hybrid maximum power point tracking technique based on PSO and incremental conductance," IEEE Conference on Energy Conversion, December 2014.